

Variational Inference

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In this report, we summarize the third section of the article¹, which provides a comprehensive example of a Bayesian mixture of Gaussians. Gaussian Mixture Models (GMMs) are a foundational concept in the applications of probabilistic modeling and clustering when one assumes that data was generated from multiple Gaussian distributions. To explicate the concepts of variational inference (VI), this section introduces a more efficient alternative method to approximate the posterior through optimization. The example previously discussed of a GMM is again presented, this time using variational methods to approximate the posterior distributions. This report presents the details of the Coordinate Ascent Variational Inference (CAVI) algorithm for Bayesian Gaussian Mixture Models, while rigorously deriving all relevant formulas.

The probabilistic model consists following components:(I) Observed data x_i represent real-valued data points. (II) Latent variables μ_k denote the mean of the k -th Gaussian cluster, and c_i represents cluster assignment for x_i .

A Bayesian Gaussian Mixture Model (GMM) is defined as follows: (I) Prior distributions for the cluster means [$\mu_k \sim \mathcal{N}(0, \sigma^2)$] (This represents a Gaussian prior centered at zero with variance σ^2). (II) Prior for the cluster assignments [$c_i \sim \text{Categorical}(\frac{1}{K}, \dots, \frac{1}{K})$]. Each data point is assigned to one of K clusters with equal probability.

Given cluster assignments, the likelihood function is defined as:

$$p(x_i|c_i, \mu) \sim \mathcal{N}(\mu_{c_i}, 1)$$

This means that data points assigned to cluster k are drawn from a Gaussian distribution with mean μ_k and unit variance. The full joint probability distribution of the observed and latent variables is defined as:

$$p(\mu, c, x) = p(\mu) \prod_{i=1}^n p(c_i) p(x_i|c_i, \mu)$$

¹arXiv:1601.00670v9 [stat.CO] 9 May 2018

Where:

- $p(\mu) = \prod_{k=1}^K \mathcal{N}(\mu_k | 0, \sigma^2)$ (Gaussian prior on means).
- $p(c_i) = \frac{1}{K}$ (Uniform prior on cluster assignments).
- $p(x_i | c_i, \mu) = \mathcal{N}(x_i | \mu_{c_i}, 1)$ (Likelihood).

Since exact posterior inference is intractable, Variational Inference (VI) approximates the true posterior $p(\mu, c | x)$ with a simpler variational distribution $q(\mu, c)$, which is chosen as:

$$q(\mu, c) = \prod_{k=1}^K q(\mu_k) \prod_{i=1}^n q(c_i)$$

Where:

- $q(\mu_k)$ is Gaussian [$q(\mu_k) = \mathcal{N}(m_k, s_k^2)$].
- $q(c_i)$ is Categorical [$q(c_i) = \text{Categorical}(\phi_i)$].

Each data point is softly assigned to clusters based on ϕ_{ik} , which represents the probability that x_i belongs to cluster k .

Definition (ELBO):² The Evidence Lower Bound (ELBO) is derived from the marginal log-likelihood:

$$\log p(x) = \log \int p(\mu, c, x) d\mu dc$$

$$\text{ELBO} = \mathbb{E}[\log p(\mu, c, x)] - \mathbb{E}[\log q(\mu, c)]$$

Use variational distribution and expanding:

$$\begin{aligned} \text{ELBO} = & \sum_{k=1}^K \mathbb{E}[\log p(\mu_k)] + \sum_{i=1}^n \mathbb{E}[\log p(c_i)] + \sum_{i=1}^n \mathbb{E}[\log p(x_i | c_i, \mu)] \\ & - \sum_{i=1}^n \mathbb{E}[\log q(c_i)] - \sum_{k=1}^K \mathbb{E}[\log q(\mu_k)] \end{aligned}$$

Each expectation term is computed separately to derive closed-form updates.

The Coordinate Ascent Variational Inference (CAVI) algorithm iteratively updates $q(c_i)$ and $q(\mu_k)$ until the Evidence Lower Bound (ELBO) converges.

²https://en.wikipedia.org/wiki/Evidence_lower_bound

For each data point x_i , the variational distribution for its cluster assignment is chosen to be a categorical distribution:

$$q(c_i) = \text{Categorical}(\phi_i)$$

Where:

$\phi_i = (\phi_{i1}, \phi_{i2}, \dots, \phi_{iK})$ is a vector of probabilities such that $\sum_{k=1}^K \phi_{ik} = 1$.

The update for $q(c_i)$ is derived by maximizing the ELBO with respect to $q(c_i)$. It can be shown that the optimal update is proportional of the exponential of the expected complete-data log likelihood respect to the other variational factors. Specifically, for each cluster k we have:

$$\phi_{ik} \propto \exp\left(\mathbb{E}_{q(\mu_k)}[\log p(x_i | c_i = k, \mu_k)] + \log p(c_i = k)\right)$$

Since the prior $p(c_i = k)$ is uniform ($\log p(c_i = k) = -\log K$), constant with respect to k , it can be absorbed into the normalization constant.

Derivation:

The likelihood for data point x_i given that it belongs to cluster k is:

$$p(x_i | c_i = k, \mu_k) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu_k)^2}{2}\right)$$

Take logarithm:

$$\log p(x_i | c_i = k, \mu_k) = -\frac{1}{2} \log(2\pi) - \frac{(x_i - \mu_k)^2}{2}$$

Expectation with Respect to $q(\mu_k) = \mathcal{N}(m_k, s_k^2)$:

$$\mathbb{E}_{q(\mu_k)}[(x_i - \mu_k)^2] = x_i^2 - 2x_i \mathbb{E}[\mu_k] + \mathbb{E}[\mu_k^2]$$

Therefore:

$$\mathbb{E}[\log p(x_i | \mu_k)] = \mathbb{E}[\mu_k] x_i - \frac{1}{2} \mathbb{E}[\mu_k^2] + \text{const}$$

And

$$\phi_{ik} \sim \exp(\mathbb{E}[(x_i \mu_k)] - \frac{1}{2} \mathbb{E}[\mu_k^2])$$

Notice it is only a function of the variational parameters for the mixture components.

We know that:

$$\mathbb{E}[\mu_k] = m_k, \quad \mathbb{E}[\mu_k^2] = m_k^2 + s_k^2$$

And the expectation of the log likelihood becomes:

$$\mathbb{E}_{q(\mu_k)}[\log p(x_i \mid c_i = k, \mu_k)] = -\frac{1}{2} \log(2\pi) - \frac{1}{2}(x_i^2 - 2x_i m_k + m_k^2 + s_k^2)$$

Discarding constant terms, we obtain:

$$\mathbb{E}_{q(\mu_k)}[\log p(x_i \mid c_i = k, \mu_k)] \propto x_i m_k - \frac{1}{2}(m_k^2 + s_k^2)$$

Thus, the optimal update for the variational parameter ϕ_{ik} becomes:

$$\phi_{ik} \propto \exp(x_i m_k - \frac{1}{2}(m_k^2 + s_k^2))$$

Finally, normalization is performed so that:

$$\phi_{ik} = \frac{\exp(m_k x_i - \frac{1}{2}(m_k^2 + s_k^2))}{\sum_{j=1}^K \exp(m_j x_i - \frac{1}{2}(m_j^2 + s_j^2))}$$

The variational distribution for the mixture component means is assumed to be Gaussian:

$$q(\mu_k) = \mathcal{N}(m_k, s_k^2)$$

The update for $q(\mu_k)$ is obtained by maximizing the ELBO with respect to $q(\mu_k)$, leading to closed-form updates.

Derivation:

Prior Contribution be:

$$p(\mu_k) = \mathcal{N}(\mu_k \mid 0, \sigma^2)$$

Taking a logarithm of the prior:

$$\log p(\mu_k) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{\mu_k^2}{2\sigma^2}$$

Likelihood Contribution

$$\prod_{i=1}^n p(x_i \mid \mu_k)^{\phi_{ik}}$$

Taking the logarithm, we have:

$$-\frac{1}{2} \sum_{i=1}^n \phi_{ik} (x_i - \mu_k)^2$$

Combine Prior and Likelihood

$$-\frac{\mu_k^2}{2\sigma^2} - \frac{1}{2} \sum_{i=1}^n \phi_{ik}(x_i - \mu_k)^2$$

Expanding the likelihood term:

$$-\frac{1}{2} \sum_{i=1}^n \phi_{ik}(x_i^2 - 2x_i\mu_k + \mu_k^2) = -\frac{1}{2} \sum_{i=1}^n \phi_{ik}x_i^2 + \sum_{i=1}^n \phi_{ik}x_i\mu_k - \frac{1}{2}\mu_k^2 \sum_{i=1}^n \phi_{ik}$$

Now, ignoring the term independent of μ_k since it does not affect the optimum, we can combine the quadratic terms:

$$-\frac{1}{2}\mu_k^2 \left(\frac{1}{\sigma^2} + \sum_{i=1}^n \phi_{ik} \right) + \mu_k \sum_{i=1}^n \phi_{ik}x_i$$

This is a quadratic function in μ_k which, when completed, gives the optimal variational parameters. The quadratic term can be written as:

$$-\frac{1}{2}A\mu_k^2 + B\mu_k$$

Where:

$$A = \frac{1}{\sigma^2} + \sum_{i=1}^n \phi_{ik}$$

$$B = \sum_{i=1}^n \phi_{ik}x_i$$

The maximum of this quadratic is achieved at:

$$m_k = \frac{B}{A} = \frac{\sum_{i=1}^n \phi_{ik}x_i}{1/\sigma^2 + \sum_{i=1}^n \phi_{ik}}$$

Moreover, because the variational distribution $q(\mu_k)$ is Gaussian, the optimal variance is the inverse of the coefficient of μ_k^2 :

$$s_k^2 = \frac{1}{A} = \frac{1}{1/\sigma^2 + \sum_{i=1}^n \phi_{ik}}$$

Conclusion

This report provides a careful derivation of Variational Inference (VI) for Bayesian Gaussian Mixture Models (GMMs). The CAVI algorithm updates posteriors with efficient closed-form updates, making Bayesian clustering scalable to large datasets.